

Question ID: 3b931fb0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Medium

Question

A right circular cylinder has a volume of **377** cubic centimeters. The area of the base of the cylinder is **13** square centimeters. What is the height, in centimeters, of the cylinder?

Correct Answer: 29

Rationale

The correct answer is **29**. The volume, V , of a right circular cylinder is given by the formula $V = \pi r^2 h$, where r is the radius of the base of the cylinder and h is the height of the cylinder. Since the base of the cylinder is a circle with radius r , the area of the base of the cylinder is πr^2 . It's given that a right circular cylinder has a volume of **377** cubic centimeters; therefore, $V = 377$. It's also given that the area of the base of the cylinder is **13** square centimeters; therefore, $\pi r^2 = 13$. Substituting **377** for V and **13** for πr^2 in the formula $V = \pi r^2 h$ yields $377 = 13h$. Dividing both sides of this equation by **13** yields $29 = h$. Therefore, the height of the cylinder, in centimeters, is **29**.